In [30]:
```
%pip install pandas numpy scikit-learn matplotlib
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.datasets import load_wine
import matplotlib.pyplot as plt
```

```
Requirement already satisfied: pandas in c:\users\alexa\appdata\local\programs\py
thon\python310\lib\site-packages (2.3.3)
Requirement already satisfied: numpy in c:\users\alexa\appdata\local\programs\pyt
hon\python310\lib\site-packages (2.2.6)
Requirement already satisfied: scikit-learn in c:\users\alexa\appdata\local\progr
ams\python\python310\lib\site-packages (1.7.2)
Requirement already satisfied: matplotlib in c:\users\alexa\appdata\local\program
s\python\python310\lib\site-packages (3.10.9)
Requirement already satisfied: tzdata>=2022.7 in c:\users\alexa\appdata\local\pro
grams\python\python310\lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: pytz>=2020.1 in c:\users\alexa\appdata\local\progr
ams\python\python310\lib\site-packages (from pandas) (2026.1.post1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\alexa\appdata\r
oaming\python\python310\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: joblib>=1.2.0 in c:\users\alexa\appdata\local\prog
rams\python\python310\lib\site-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\alexa\appdata\loc
al\programs\python\python310\lib\site-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: scipy>=1.8.0 in c:\users\alexa\appdata\local\progr
ams\python\python310\lib\site-packages (from scikit-learn) (1.15.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\alexa\appdata\local\p
rograms\python\python310\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: pillow>=8 in c:\users\alexa\appdata\local\programs
\python\python310\lib\site-packages (from matplotlib) (12.2.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\alexa\appdata\local
\programs\python\python310\lib\site-packages (from matplotlib) (4.62.1)
Requirement already satisfied: packaging>=20.0 in c:\users\alexa\appdata\roaming
\python\python310\site-packages (from matplotlib) (26.2)
Requirement already satisfied: cycler>=0.10 in c:\users\alexa\appdata\local\progr
ams\python\python310\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\alexa\appdata\local
\programs\python\python310\lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: pyparsing>=3 in c:\users\alexa\appdata\local\progr
ams\python\python310\lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in c:\users\alexa\appdata\roaming\python
\python310\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip available: 22.2.2 -> 26.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [31]:
```
wine = load_wine()

df = pd.DataFrame(wine.data, columns=wine.feature_names)
df['target'] = wine.target

df.head()
```

Out[31]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nc
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	

In [32]:

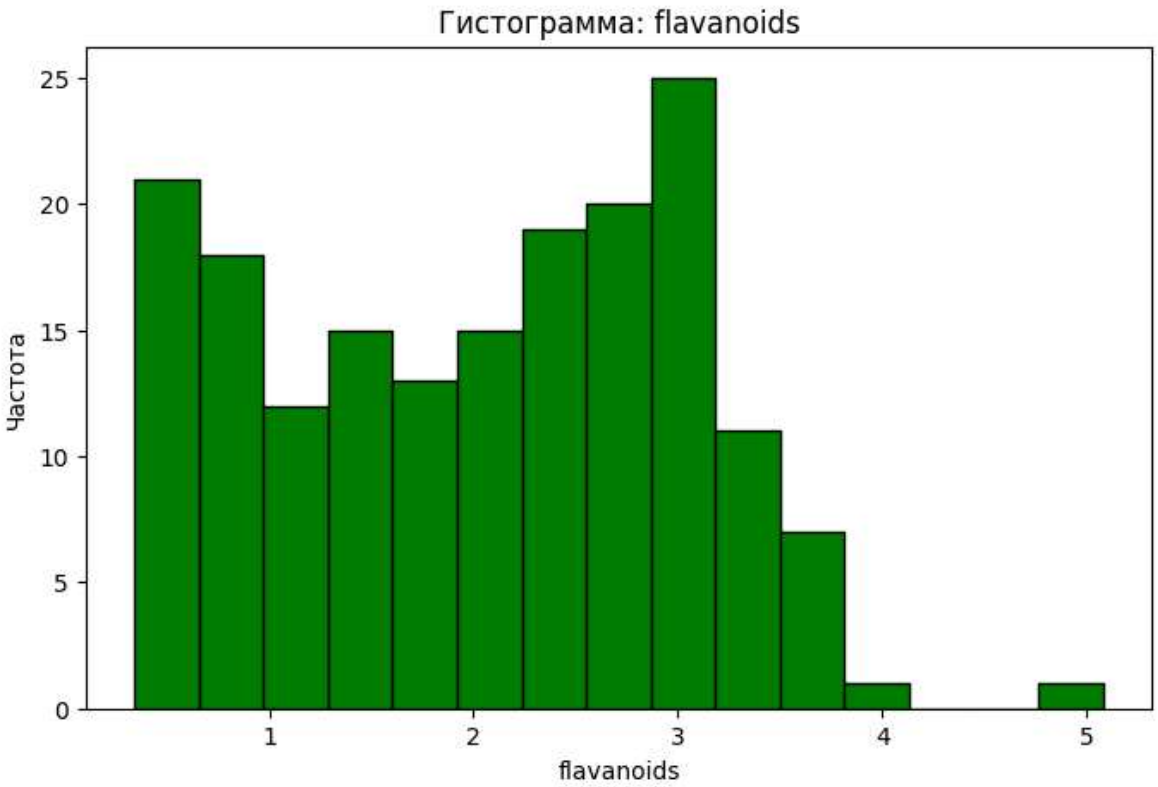
```
print(f"Форма датасета: {df.shape}")
print(f"\nНазвания признаков:\n{wine.feature_names}")
print(f"\nНазвания классов: {wine.target_names}")
print(f"\nРаспределение классов:\n{df['target'].value_counts()}")
print(f"\nСтатистика:\n{df.describe()}")
```

```
Форма датасета: (178, 14)

Названия признаков:
['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'total_phenol
s', 'flavanoids', 'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity',
'hue', 'od280/od315_of_diluted_wines', 'proline']

Названия классов: ['class_0' 'class_1' 'class_2']

Распределение классов:
target
1    71
0    59
2    48
Name: count, dtype: int64

Статистика:
          alcohol  malic_acid         ash  alcalinity_of_ash   magnesium  \
count  178.000000  178.000000  178.000000         178.000000  178.000000
mean    13.000618    2.336348    2.366517          19.494944   99.741573
std      0.811827    1.117146    0.274344           3.339564   14.282484
min     11.030000    0.740000    1.360000          10.600000   70.000000
25%     12.362500    1.602500    2.210000          17.200000   88.000000
50%     13.050000    1.865000    2.360000          19.500000   98.000000
75%     13.677500    3.082500    2.557500          21.500000  107.000000
max     14.830000    5.800000    3.230000          30.000000  162.000000

       total_phenols  flavanoids  nonflavanoid_phenols  proanthocyanins  \
count     178.000000  178.000000            178.000000       178.000000
mean        2.295112    2.029270              0.361854         1.590899
std         0.625851    0.998859              0.124453         0.572359
min         0.980000    0.340000              0.130000         0.410000
25%         1.742500    1.205000              0.270000         1.250000
50%         2.355000    2.135000              0.340000         1.555000
75%         2.800000    2.875000              0.437500         1.950000
max         3.880000    5.080000              0.660000         3.580000

       color_intensity         hue  od280/od315_of_diluted_wines      proline  \
count       178.000000  178.000000                    178.000000   178.000000
mean          5.058090    0.957449                      2.611685   746.893258
std           2.318286    0.228572                      0.709990   314.907474
min           1.280000    0.480000                      1.270000   278.000000
25%           3.220000    0.782500                      1.937500   500.500000
50%           4.690000    0.965000                      2.780000   673.500000
75%           6.200000    1.120000                      3.170000   985.000000
max          13.000000    1.710000                      4.000000  1680.000000

           target
count  178.000000
mean     0.938202
std      0.775035
min      0.000000
25%      0.000000
50%      1.000000
75%      2.000000
max      2.000000
```

In [33]:

```
column_for_hist = 'flavanoids'
plt.figure(figsize=(8, 5))
plt.hist(df[column_for_hist], bins=15, color='green', edgecolor='black')
```

```
plt.xlabel(column_for_hist)
plt.ylabel('Частота')
plt.title(f'Гистограмма: {column_for_hist}')
plt.show()
```



In [34]:
```
# Создание категориального признака на основе alcohol
df['alcohol_category'] = pd.cut(df['alcohol'],
                                bins=3,
                                labels=['low', 'medium', 'high'])

print("Уникальные значения нового категориального признака:")
print(df['alcohol_category'].unique())
print("\nРаспределение по категориям:")
print(df['alcohol_category'].value_counts())

df.head()
```

```
Уникальные значения нового категориального признака:
['high', 'medium', 'low']
Categories (3, object): ['low' < 'medium' < 'high']

Распределение по категориям:
alcohol_category
medium    86
high      51
low       41
Name: count, dtype: int64
```

Out[34]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nc
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	

OneHotEncoding

In [35]:

```
df_ohe = df.copy()

ohe = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
ohe_array = ohe.fit_transform(df_ohe[['alcohol_category']])

ohe_array_df = pd.DataFrame(
    ohe_array,
    columns=ohe.get_feature_names_out(['alcohol_category']),
    index=df_ohe.index
)

df_ohe_final = pd.concat([df_ohe, ohe_array_df], axis=1)
df_ohe_final.head()
```

Out[35]:

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	flavanoids	nc
0	14.23	1.71	2.43	15.6	127.0	2.80	3.06	
1	13.20	1.78	2.14	11.2	100.0	2.65	2.76	
2	13.16	2.36	2.67	18.6	101.0	2.80	3.24	
3	14.37	1.95	2.50	16.8	113.0	3.85	3.49	
4	13.24	2.59	2.87	21.0	118.0	2.80	2.69	

Label Encoding

In [36]:

```
label_mapper = {
    'low': 0,
    'medium': 1,
    'high': 2
}

df['alcohol_category_label'] = df['alcohol_category'].map(label_mapper)

print("Результат Label Encoding:")
df[['alcohol', 'alcohol_category', 'alcohol_category_label']].head()
```

```
Результат Label Encoding:
```

Out[36]:

	alcohol	alcohol_category	alcohol_category_label
0	14.23	high	2
1	13.20	medium	1
2	13.16	medium	1
3	14.37	high	2
4	13.24	medium	1

In [37]:

```
scaler = StandardScaler()
df['flavanoids_scaled'] = scaler.fit_transform(df[['flavanoids']])

print("Результат стандартизации:")
df[['flavanoids', 'flavanoids_scaled']].head()
```

```
Результат стандартизации:
```

Out[37]:

	flavanoids	flavanoids_scaled
0	3.06	1.034819
1	2.76	0.733629
2	3.24	1.215533
3	3.49	1.466525
4	2.69	0.663351

In [38]:

```
print("Статистика исходного признака flavanoids:")
print(df['flavanoids'].describe())
print("\nСтатистика scaled признака:")
print(df['flavanoids_scaled'].describe())
```

```
Статистика исходного признака flavanoids:
count    178.000000
mean       2.029270
std        0.998859
min        0.340000
25%        1.205000
50%        2.135000
75%        2.875000
max        5.080000
Name: flavanoids, dtype: float64

Статистика scaled признака:
count    1.780000e+02
mean    -3.991813e-16
std      1.002821e+00
min     -1.695971e+00
25%     -8.275393e-01
50%      1.061497e-01
75%      8.490851e-01
max      3.062832e+00
Name: flavanoids_scaled, dtype: float64
```

In [39]:

```
columns_to_show = ['alcohol', 'alcohol_category', 'alcohol_category_label',
                   'flavanoids', 'flavanoids_scaled', 'target']
```

```
print("Итоговая таблица (первые 10 строк):")
df[columns_to_show].head(10)
```

```
Итоговая таблица (первые 10 строк):
```

Out[39]:

	alcohol	alcohol_category	alcohol_category_label	flavanoids	flavanoids_scaled	targe
0	14.23	high	2	3.06	1.034819	
1	13.20	medium	1	2.76	0.733629	
2	13.16	medium	1	3.24	1.215533	
3	14.37	high	2	3.49	1.466525	
4	13.24	medium	1	2.69	0.663351	
5	14.20	high	2	3.39	1.366128	
6	14.39	high	2	2.52	0.492677	
7	14.06	high	2	2.51	0.482637	
8	14.83	high	2	2.98	0.954502	
9	13.86	high	2	3.15	1.125176	